Project Documentation (Word File)

Formatting Rules

- **Font Style:** *Times New Roman*
- **Font Size:**
  - **12** for all normal content
  - **16** for all **Sub_Headings**
  - **20 for all heading**
- **Page Numbers:** Add at the bottom

2.Compulsory Sections in Documentation

1. Preliminary Investigation

1.1 Problem Identification

Problem Identification explains **why the project is undertaken**.
It highlights the real-world problems, difficulties, or inefficiencies that inspired the development of the system.

Example:
The problem Identification behind this project is to reduce manual effort, improve accuracy, save time, and provide an efficient computerized solution to manage data effectively.

1.2 Problem Statement/Definition

This clearly defines **the exact problem** that the project aims to solve.
It should be precise and easy to understand.

Example:
The existing system is manual, time-consuming, error-prone, and lacks proper data security. Hence, there is a need for an automated system.

1.3 Purpose / Objective and Goals

This section explains **what the project intends to achieve**.

Objectives:

- To automate the existing system
- To reduce human errors
- To improve data processing speed
- To ensure data security and reliability

Goals:

- Develop a user-friendly system
- Provide accurate and quick results
- Improve overall system efficiency

1.4 Feasibility Study

You can include these 4 types:

1. **Technical Feasibility**
    - Whether the required technology is available.
    - Programming language, database, software, hardware, etc.
    - Example: *The system can be developed using Python, HTML/CSS, JavaScript and MySQL.*
2. **Economic Feasibility**
    - Whether the project is affordable.
    - Cost of software, hardware, hosting, internet, etc.
    - Mention if you are using free/open-source tools, making the project low-cost.
3. **Operational Feasibility**
    - Whether the proposed system will be easy and useful for users.
    - How it will improve the existing process.
    - Example: *The system reduces manual work and makes information retrieval faster.*
4. **Schedule Feasibility**

- Whether you can complete the project within your college deadline.
- You can mention stages such as:
  - Requirement gathering
  - System design
  - Coding
  - Testing
  - Documentation
  - Final implementation

1.5 Project Scope and Limitations

Project Scope:

- Defines what the system **will do**
- Identifies features and functionalities

Example:

- Data entry and management
- User authentication
- Report generation

Limitations:

- Internet dependency (if web-based)
- Limited to predefined users
- Does not support advanced analytics

---

2. Requirement Specification

2.1 System Requirements

Software/Hardware specifications – (if required) – specification of server / device Platform, RAM, Memory storage

2.2 Technical Requirement – Programming Language, Tools if any

---

2.3 Functional Requirements

functional requirements describe what the system should do—the features, actions, and services provided by the system.

Typically, you can include:

1. User Registration
    - User should be able to create an account.
    - System should store user details.
2. Login / Logout
    - User should be able to log in using valid credentials.
    - User should be able to log out securely.
3. User Management
    - Admin should be able to add, update, delete, or view users.
4. Data Entry
    - Users should be able to enter required information into the system.
    - System should validate the entered data.
5. Data Management
    - System should allow authorized users to add, view, update, and delete records.
6. Search
    - Users should be able to search for required information using relevant keywords or filters.
7. Reports
    - System should generate reports based on stored data.
    - Reports may be viewed, downloaded, or printed.
8. Notifications
    - System should provide notifications or alerts when required.
9. Admin Functions
    - Admin should be able to manage users, records, settings, and other system operations.
10. Security / Access Control

- Different users should have appropriate permissions.
- Unauthorized users should not be able to access restricted features.

2.4 Data Requirements

(As per the project requirement, student can include – Performance requirement & security requirements etc.)

---

3. Database Design

3.1 Identify End Users of the System

- Types of users
- User roles
- User permissions
- Activities performed by each user

3.2 Identify Entities and Attributes through ER Diagram

- List of entities
- Attributes of each entity
- Primary keys
- ER diagram
- Explanation of entities and relationships
- Cardinality

3.3 Identify All Tables, Fields, Relationships Between Tables

- List of database tables
- Table structure
- Field/column names
- Data types
- Primary keys
- Foreign keys
- Description of fields
- Relationships between tables

- Relationship type (1:1, 1:M, M:N)

4.System Design

contains UML diagrams that explain how your system is structured and how it works.

4.1 Class Diagram

A Class Diagram shows the classes in your system, their attributes, methods/functions, and relationships.

4.2 Object Diagram

An Object Diagram shows actual instances/objects of classes at a particular point in time.

4.3 Component Diagram

A Component Diagram shows the major software components/modules of your system and how they interact.

4.4 Deployment Diagram

A Deployment Diagram shows where the software components are physically deployed/running.

4.5 Use Case Diagram

A Use Case Diagram shows what different users can do with the system.

4.6 Activity Diagram

An Activity Diagram shows the step-by-step workflow of a process.

4.7 Sequence Diagram

A Sequence Diagram shows how different users/objects/components communicate with each other in a particular sequence over time**.**

5. Input and Output Screen Layout

5.1.1 Login Screen

- Purpose
- Screen design/screenshot
- Description of fields and buttons

5.1.2 Registration Screen

- Purpose
- Screen design/screenshot
- Description

5.1.3 Add Record Screen

- Purpose
- Screen design/screenshot
- Description

5.2 Output Screen Layout

5.2.1 Dashboard

- Purpose
- Screenshot
- Description

5.2.2 Search Result Screen

- Purpose
- Screenshot
- Description

5.2.3 Report Screen

- Purpose
- Screenshot
- Description

6.Bibliography / References

Bibliography / References section lists all the sources you used while developing and documenting your project.

These can include:

- Books – textbooks or reference books used for programming, database, software engineering, etc.
- Websites – official documentation and educational websites.
- Research papers/articles – if you referred to any.
- Software documentation – documentation for programming languages, frameworks, databases, APIs, etc.
- Tutorials – online learning resources used for understanding implementation.
- Images/resources – if you used diagrams, icons, or other external resources.

Teacher Review

- After completing your project Word document, show your respective assign teacher for
- review.
- The teacher will check your formatting, content, diagrams, and overall presentation.
- Make all the improvements suggested by your teacher.
- After applying the changes,prepare the final version of your documentation for
- submission.